

Microgrid Design and Operation

Project 2, due 4/26

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Project 2 objective: Build a microgrid to satisfy power needs for critical services in an island. The island is Oahu, Hawaii. Project 2 focuses on power engineering, whereas Project 1 focused on economic optimization.

Problems for all students (80 points)

Task 1: From the optimal system size in Project 1 Task 3, provide screenshots of the inputs of the load profile and all assets. **(5 pts)**

The image displays two screenshots of a microgrid design software interface, showing the configuration of a Diesel Generator and a Controller.

Diesel Generator Configuration:

- Asset Name:** Diesel Generator
- Technology Decision:** Consider, Existing, Existing + More, Forced, Forced + More (Existing is selected)
- Existing Number:** Existing Number (dropdown)
- Existing Age:** Existing Age (dropdown)
- Max Number:** 30
- Investment Specifications:**
 - Per Unit Install Cost:** \$ 85,000
 - Lifetime:** 15 Years
 - Variable Maintenance:** \$ 0.20 /kWh output
 - Annual Fixed Maintenance:** \$ 35 /kWh/Year
 - Runtime Maintenance:** \$ 1.40 /Hour/Unit
- Technical Specifications:**
 - Rated Size:** 100 kW
 - Nameplate Efficiency:** 30 %
 - Temperature Derating:** Temperature Derating Curve (dropdown)
 - Fuel Type:** Diesel Fuel
 - Backup Only:** YES
- Incentives:**
 - Investment Tax Credit:** 0 %
 - Amount Depreciable:** 0 %
 - MACRS Years:** N/A
 - Production Tax Credit:** \$ 0 /kWh

Controller Configuration:

- Manufacturer:** Controller
- Lifetime:** 15 Years
- Investment Specifications:**
 - Install Cost:** \$ 155,000
 - Annual Support:** \$ /Year
 - Other:** \$

Menu

Decisions

Financial

System Info

Catalog

Save to Catalog

Manage

Wind Turbine

Existing Name

Custom

Technology Decision

Consider

Existing

Existing + More

Forced

Forced + More

Existing Number

Existing Age

Max Number

Investment Specifications

Per Unit Install Cost

Lifetime

Variable Maintenance

Monthly Fixed Maintenance

Incentives

Investment Tax Credit

Amount Depreciable

MACRS Years

Production Tax Credit

REC

Technical Specifications

Upload Wind Performance Data

Model

Rated Size

Hub Height

Power Factor

Curtailable

Electricity Export

Compute

Save

Cancel

Delete

Menu

Decisions

Financial

Incentives

System

Catalog

Save to Catalog

Manage

Solar PV

Existing Name

Custom

Technology Decision

Consider

Existing

Existing + More

Forced

Forced + More

Existing Size

Existing Age

Max New Size

Investment Specifications

Inverter

Inverter Cost

Lifetime

Power Factor

Custom Inverter

Solar PV

Per Unit Install Cost

Lifetime

Variable Maintenance

Monthly Fixed Maintenance

Fixed Install Fees

Define Space

Install Space

Unit Spacing

Panel Rating

Incentives

Investment Tax Credit

Amount Depreciable

MACRS Years

REC

Electricity Export

Production Tax Credit

Save

Cancel

Delete

Electric Utility Tariff

Tariff Name

Schedule J - General Service Demand

This tariff applies to any building load plus all technologies except for EV Chargers that are explicitly assigned to an alternative electric tariff.

Analysis Optimization #8 is running. We'll notify you when it completes.

Energy Charges

Demand Charges

Fees

Exports

Season Definition

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

To add another season, click on any of the months to get a dropdown list and select the season that applies.

Contracted Tariff

Energy Charge Description

Add Tier

Add TOU Period

Season 1

Period 1

\$ 0.053177 / kWh

Real Time Price

Upload RTP Data

Upload Customer Baseline Load Data

Electric Utility Tariff

Tariff Name

Schedule J - General Service Demand

This tariff applies to any building load plus all technologies except for EV Chargers that are explicitly assigned to an alternative electric tariff.

Analysis Optimization #8 is running. We'll notify you when it completes.

Energy Charges

Demand Charges

Fees

Exports

Season Definition

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Season 1

Demand [\$ / kW]

Monthly

To add another season, click on any of the months to get a dropdown list and select the season that applies.

Contracted Tariff

Demand Charge Description

Add Coincident Hour

Add Tier

Add TOU Period

Season 1

Non-Coincident Rate

\$ 13 / kW

Show Demand Constraints

Electric Utility Tariff

Tariff Name

Schedule J - General Service Demand

This tariff applies to any building load plus all technologies except for EV Chargers that are explicitly assigned to an alternative electric tariff.

Energy Charges

Demand Charges

Fees

Exports

Fees

Access Fee

\$

98.2

/ Month

Contract Demand Fee

\$

0

/ kW

Standby Charge

\$

0

/ kW/Month

Electric Utility Tariff

Tariff Name

Schedule J - General Service Demand

This tariff applies to any building load plus all technologies except for EV Chargers that are explicitly assigned to an alternative electric tariff.

Analysis Options

We'll notify you

Energy Charges

Demand Charges

Fees

Exports

Exports

Sales Price

Sales Price Same As Tariff

Net Metering Constraint

YES

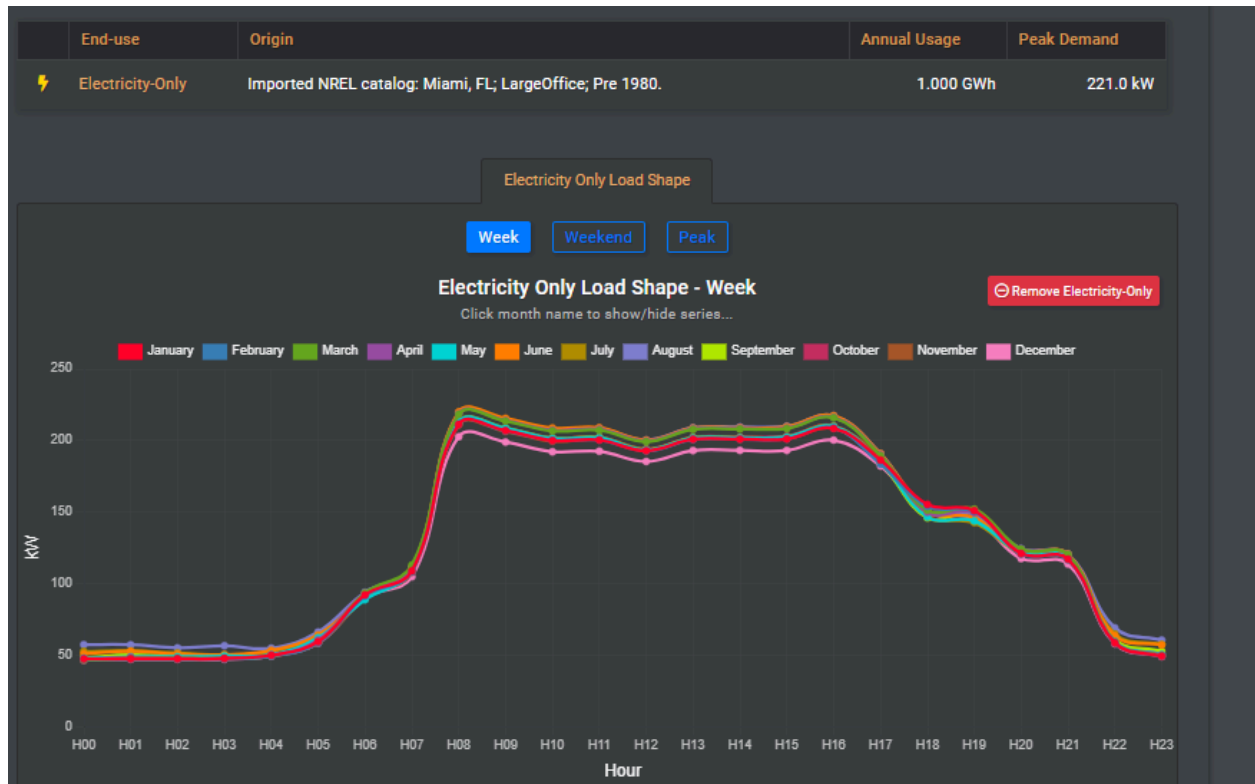
Net Metering Limit

100 %

Contracted Sales Price

Rates For electricity are equal to the rates set for Energy Charges and apply at the same TOU periods defined for Energy Rates. For tiered Energy Rates, the Tier 1 rates are used to set the sales price.

Show Export Constraints



Diesel Fuel Purchases

Monthly Access Fee

\$

Fee

Monthly

\$

/Gallon

Emissions Factor

0.24964

kgCO₂/kWh

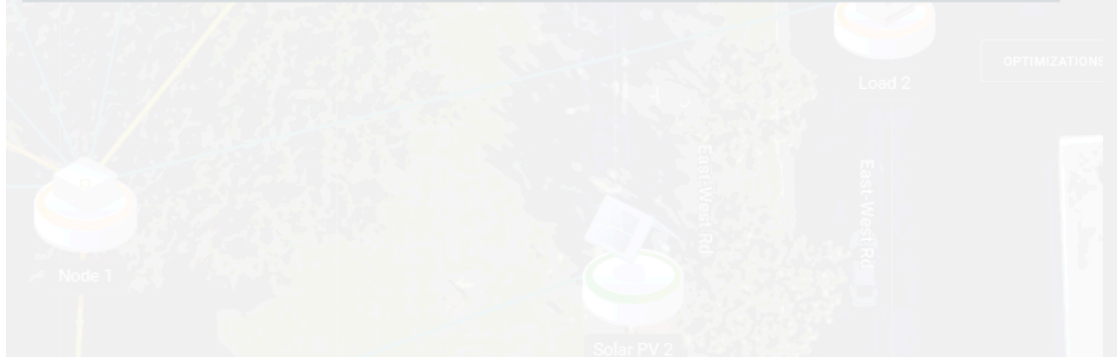
Annual Limit

Gallons

Every asset tab and load profile was provided above.

Task 2: From the optimal system size in Project 1 Task 3, provide a table of the optimal asset sizes. (5 pts)

Type	Total New Capacity	Technology (New Capacity)	Init. Equip. Cost ?
	2.58 MW	Node 2: Solar PV 2 (2.58 MW)	4,521 k\$
	213 kWh	Node 1: Battery Energy Storage System (213 kWh)	91 k\$
	1.1 MW	Node 1: Wind Turbine (1.1 MW) (11 units)	3,300 k\$
Total Cost: 7,912 k\$			



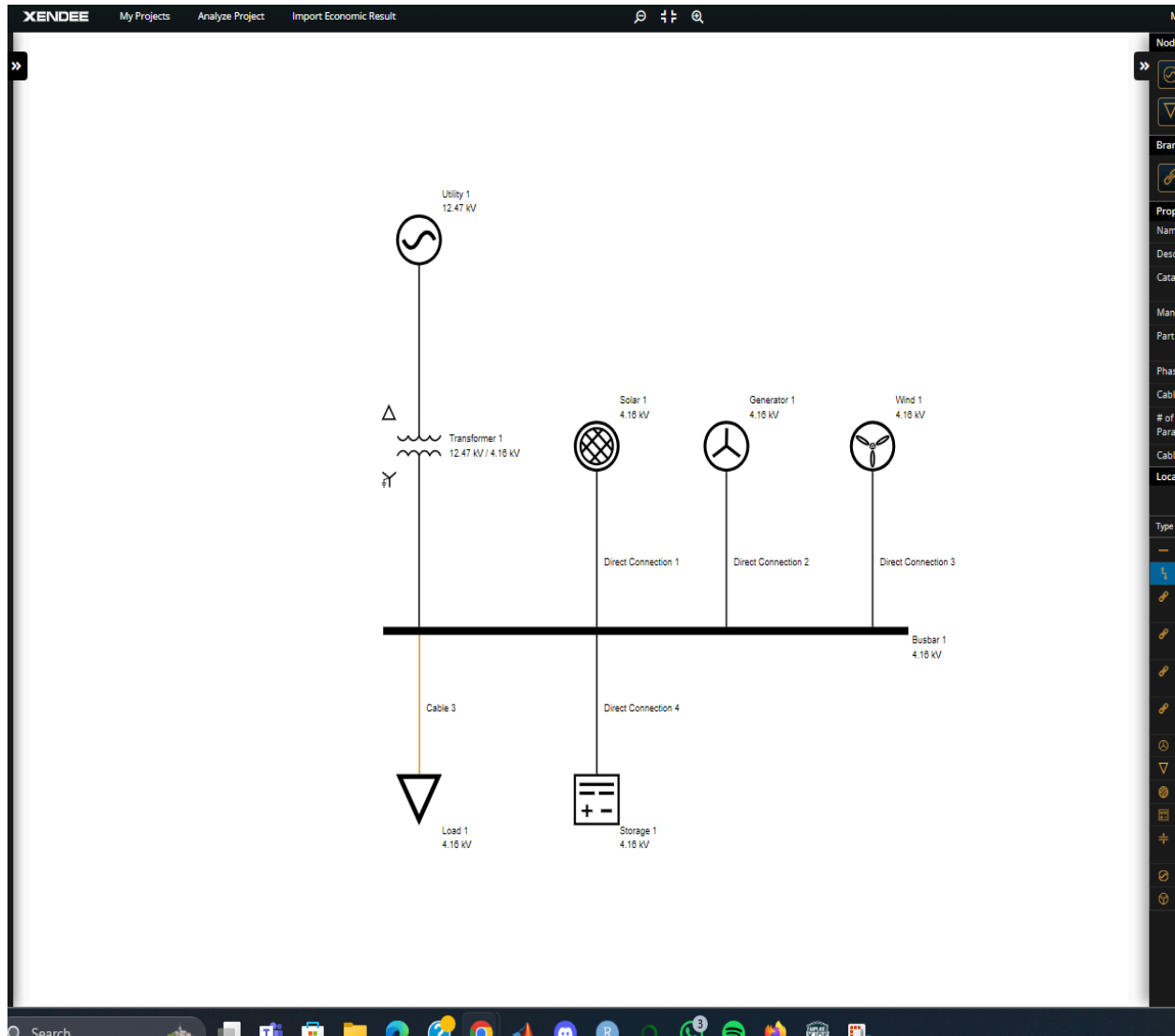
Type	Omitted Technology, Unit Size
	Node 1: Diesel Generator, 100 kW
	Node 3: Solar PV, N/A

Task 3: Draw a one-line schematic of the system (10 pts)

- A 12.47 kV AC distribution feeder
- A 480 V AC main bus
- A transformer connecting the distribution feeder to the 480 VAC main bus (use delta to wye-solidly transformer), and add an isolation breaker on each side of the transformer
- Connect the load using one of two approaches:
 - If peak load is ≥ 250 kW, then connect the load to the main 480V bus through a 100 ft cable
 - If peak load is < 250 kW, then use a step-down transformer to a 240V node (use wye-solidly to wye-solidly transformer), and then use a 100 ft cable to connect the 240V node to the load
- And a microgrid node and connect it to the main bus through a 300 ft cable, keep the main microgrid node at 480 VAC, and be sure to add inverters when necessary for any DC sources connected to the main microgrid AC bus
- Be sure to label the voltage of all buses

Task 4: Recreate Task 3 in Xendee One-line. Be sure to input any of the asset sizing and technical parameters from Project 1. Use the peak load to calculate minimum requirements for transformer and cable sizing using a 1.25x oversizing requirement (for transformer sizing in kVA, recall the relationship $kW = kVA * PF$, you have the kW of the peak load and the power factor needed to calculate kVA) (30 pts)

a) Provide a screenshot of the one-line schematic system



b) Provide a screenshot of input data windows from all assets and balance of system

Utility 1

Name *	Utility 1	Description		Default Utility	Catalog
Voltage (L - L) *	12.47 kV	Operating Voltage *	12.47 kV	3-Phase Short Circuit *	100000 kVA
Voltage Angle	0 degrees	X/R (+)	5.25	Line-Ground Short Circuit	100000 kVA
Base MVA	100 MVA	X/R (0)	5.25	Stiffness	on
Resistance (+)	0.1871121 PU	Resistance (0)	0.1871121 PU		
Reactance (+)	0.9823386 PU	Reactance (0)	0.9823386 PU		

Save + Catalog Cancel Save

Direct Connection 1 Direct Connection 2 Direct Connection 3

Busbar 1 4.16 kV

Diagram for Question 10: A series circuit with a battery (E), a switch (S), and a resistor (R). The switch is open.

Direct Connection 4

Cable 3

Name *	Length (feet)	1/0 ALGUARD Quadruplexe...
Cable 3	100 Override	Catalog
Description	# of Cables in Parallel	Size (Non-Magnetic)
	1 Change	1/0 AWG/kcmil
Cable Material	From Side Connection	Protection
Aluminum ▼	— Busbar 1	none ▼
Ampacity	To Side Connection	Protection
0.198 kA	▽ Load 1	none ▼

[Save + Catalog](#) [Cancel](#) [Save](#)

12.47 kV / 4.16 kV

Direct Connection 1 Direct Connection 2 Direct Connection 3

Busbar 1 4.16 kV

Cable 3 Direct Connection 4

Report Economic Result

Load 1

Name * Load 1 Description Default Load Catalog

Nominal Voltage (L - L) 4.16 kV Rated Voltage * 4.16 kV Rated Power * 221.722 kW

Power Factor 55 % Power Factor Type Lag

Minimum Voltage 0.75 PU Load Model

- Constant kVA
- Constant Current
- Constant Impedance
- Constant P Quadratic Q
- Linear P Quadratic Q

Connection Wye-Solidly

EV Charging Load off

Save + Catalog Cancel Save

Direct Connection 1 Direct Connection 2 Direct Connection 3

Busbar 1 4.16 kV

Cable 3 Direct Connection 4

Report Economic Result

Solar 1

Name * Solar 1 Description Default Solar-PV Catalog

Voltage (L - L) 4.16 kV Hold Voltage on Force Seq (+) on

Rated Power * 2580 kW Power Factor 100 %

Connection Wye-Solidly X/R 0.03 Rating 0.3581 kA

Inverter Rating 2580 kVA Fault (x) 4 Impedance 25 %

Install Space 99999999 ft² Cut-In Power 10 % Cut-Out Power 10 % Isc Max 1.4323 kA

Save + Catalog Cancel Save

Direct Connection 1 Direct Connection 2 Direct Connection 3

Generator 1

Name *	Description	CAT D80-100 (100kW)
Generator 1		Catalog
Voltage (L - L)	Fuel Type	Rated Power *
4.16 kV	Diesel	100 kW
Rated RPM	Poles	
1800 RPM	4	
Power Factor	Connection	
80 %	Wye-Solidly	

Save + Catalog Cancel Save

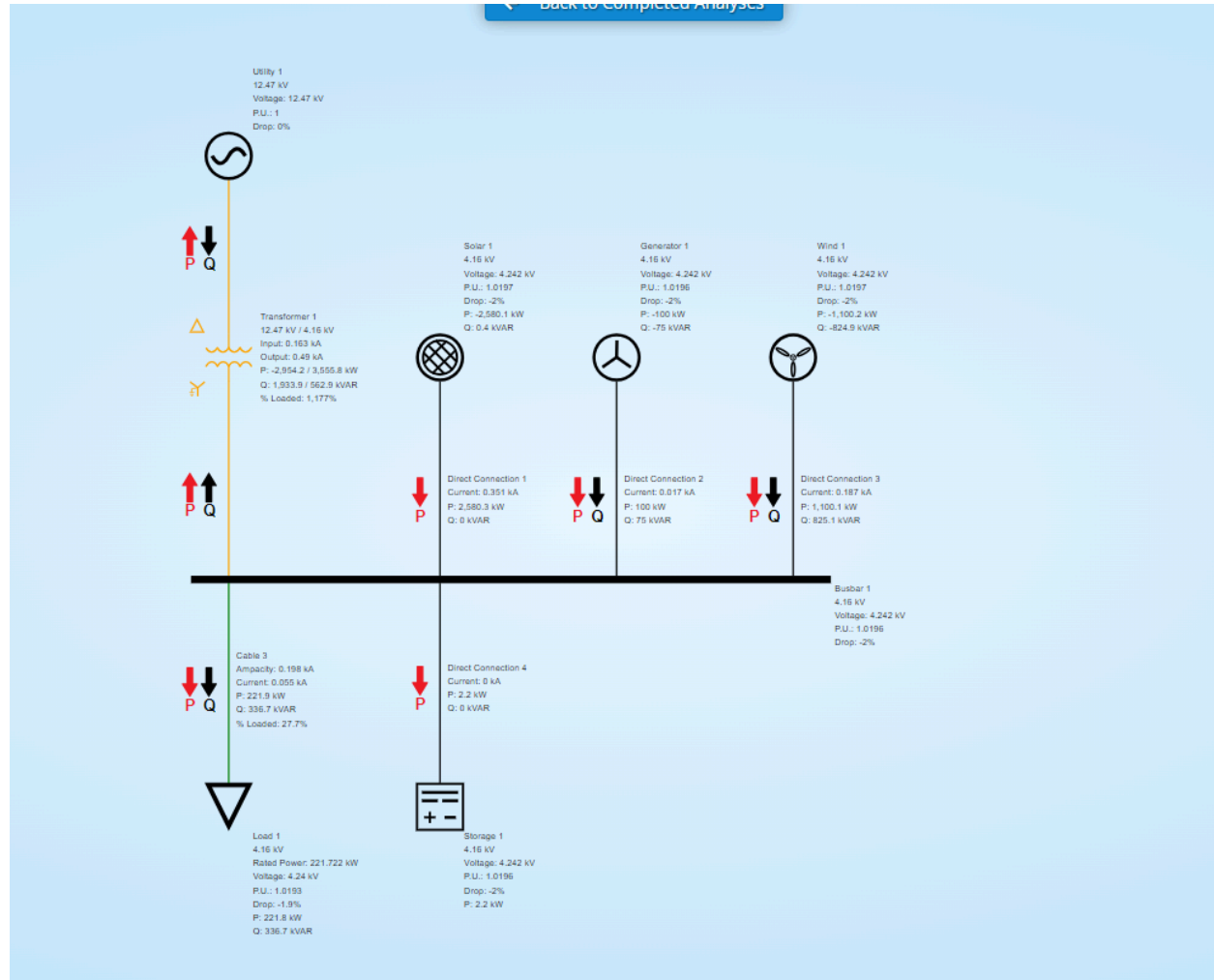
12.47 kV / 4.16 kV

Direct Connection 1 Direct Connection 2 Direct Connection 3

Busbar 1
4.16 kV

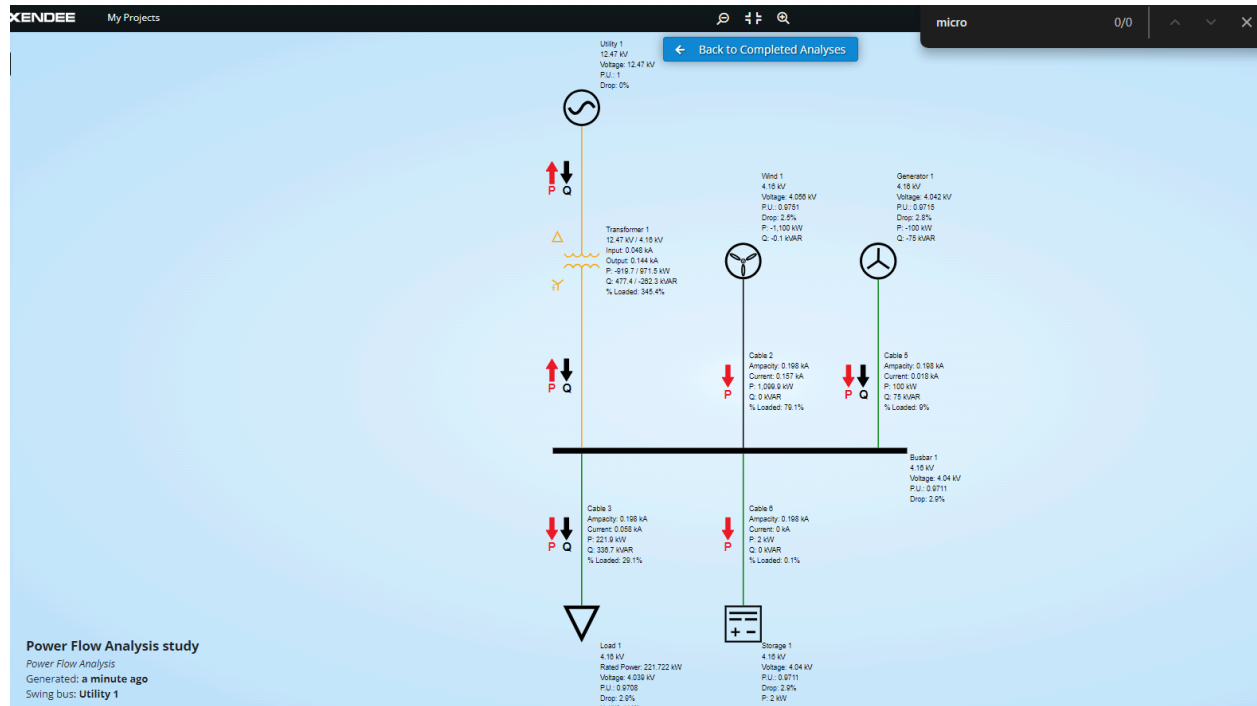
Cable 3 Direct Connection 4

Task 5: Run a power flow analysis for the PEAK load. Be sure to correct any current or voltage violations. Provide a screenshot of the power flow results. (10 pts)



Task 6: Disconnect from the grid. Update settings of the microgrid assets if needed. Run a power flow analysis for the peak load. Be sure to correct any current or voltage violations. Check to be sure your assets can provide sufficient power and aren't being operated in excess of their power output, if they are, then use assets from your 3-day resilience analysis and rerun until solution is feasible (10 pts)

Disconnections established. Results are below and the microgrid updates assets as well.



Analyze Project Import Economic Result

micro 0/0

Wind 1

Name *	Description	Default Wind
Wind 1		Catalog

Voltage (L - L)	Rated Power *
4.16 kV	1100 kW

WTG Type	Power Factor Full Load	Shunt Capacitor Stages
☒ (1) Squirrel-Cage Induction	100 %	
☐ (2) Wound Rotor Induction	Power Factor Correction	Shunt Capacitor Rating
☐ (3) Doubly-Fed Asynchronous		
☐ (4) Full Converter Interface		

Steady State	Transient	Max Power Delivery
%	%	☐ 300000 MVAR
Subtransient X0		Max Power Absorption
%	%	☐ -300000 MVAR
X/R	X2	
	%	

Turbine Model	Hub Height



Task 7: Use 3-5 sentences to compare results from Task 5 and 6, what is being provided by the grid or assets, what is happening with real and reactive power flow, what is happening with voltage levels? (10 pts)

Comparing the results from Tasks 5 and 6, it's evident that the operational dynamics of the microgrid vary significantly between grid-connected and islanded modes. In Task 5, the microgrid benefits from the stability and backup of the grid, receiving supplemental power to manage real and reactive power flows effectively, and maintain stable voltage levels. In contrast, Task 6 showcases the

microgrid's autonomy, with local generation assets—solar panels, wind turbines, and a diesel generator—shouldering the entire load independently. This shift necessitates a careful balance of power generation and storage to prevent voltage fluctuations and ensure the demands are met. Moreover, the change from grid support to island mode requires the microgrid to rely heavily on its control systems and inverters to regulate power quality and distribution. This underscores the critical nature of asset sizing and management, ensuring that each component is capable of operating within its limits while collectively supporting the microgrid's load requirements, particularly during peak demand.